Semester Project: Domain Discovery, Recommendation, and Graph Intelligence

Milestone: Recommendation, Ranking, and Predictive Decision Engine (Week 10)

Iam Anthony Marcelo Alvarez Orellana | Jeffrey Ulises Diaz Villanueva
Paula Jimena Mancilla Cienfuegos | Fernando Samuel Paredes Espinoza

1. Task Framing

Task type: Recommendation.

Given a user's positive interaction history (games endorsed via voted_up=True), the system predicts which unseen game they would also endorse. Output: ranked list.

Clustering results from Week 7 (6 segments: Casual, Action, Local Multiplayer, Narrative/JRPG, VR, Strategy) feed into this pipeline as candidate-filtering features.

2. Data Alignment

All systems operate on the same 500-game universe from data/processed/steam_v1.parquet. AppID is the common join key:

- steam_v1.parquet - user records (author_steamid, AppID, voted_up, playtime)
- games_may2024_full.csv - game metadata (name, tags, genres, categories)
- artifacts/clustering/cluster_labels_k6.npy - cluster labels from Week 7

3. Evaluation Protocol

- Interaction matrix: sparse binary, users x games. voted_up=True -> 1.
- Qualifying users: >= 2 distinct positive interactions.
- Train/test split: one positive per user held out (random_state=42).
- Primary metric: HR@10 - fraction of users whose held-out game is in top-10.
- Leakage prevention: training items masked to -inf before ranking.

Statistic	Value
Positive interactions	442,652
Unique users	426,601
Unique games (universe)	484
Qualifying users (eval)	12,389
Matrix density	0.0078%

4. Baseline 1 - Popularity

Design: All games ranked by aggregate 'recommendations' count. Same ranking for every user.
Justification: Non-personalized ceiling. Any personalized system failing to beat it is unjustified.
Limitation: Biases toward AAA titles; cannot surface niche Indie games.

5. Baseline 2 - Content-Based

TF-IDF on name + description + tags + genres + categories (1-2 grams, 3,000 features). TruncatedSVD -> 50D. User vector = centroid of training game vectors. Ranking by cosine similarity.
Limitation: Overspecializes to known preferences; ignores collective behavioral signal.

6. System 3 - Matrix Factorization (SGD)

Objective:
  min_{P,Q} sum_{(u,i)} (R_ui - p_u^T q_i)^2 + lambda*(||P||^2 + ||Q||^2)

SGD updates per pair (u,i):
  p_u <- p_u + eta*(e_ui*q_i - lambda*p_u)
  q_i <- q_i + eta*(e_ui*p_u - lambda*q_i)
  where e_ui = R_ui - p_u^T q_i

Parameter	Value	Justification
Learning rate eta	0.03	Standard for implicit binary data
Regularization lambda	0.01	Prevents overfitting on 99.8% sparse matrix
Epochs	30	Sufficient for convergence in 500-game universe
Latent dim k	{10,20,50}	Swept via HR@10; best k=10 used for final model
Init	N(0,0.01)	Small random values to break symmetry

7. System 4 - Hybrid Recommender

s_hybrid(u,g) = alpha * s_content(u,g) + (1-alpha) * s_MF(u,g)
Both scores min-max normalized to [0,1]. alpha in {0.1, 0.3, 0.5, 0.7, 0.9}.

Hybrid is legitimate: AppID is the common key for both content matrix and MF item factors. Both systems index games identically - no cross-catalog alignment issue.

8. Offline Evaluation Results

Generated by python src/recommendation.py with random_state=42.

System	HR@5	HR@10	HR@20
Popularity baseline	0.4597	0.5796	0.7124
Content-based baseline	0.1806	0.2170	0.2669
MF k=10	0.1103	0.1888	0.2949
MF k=20	0.0764	0.1333	0.2183
MF k=50	0.0756	0.1235	0.1998
Hybrid (alpha=0.3, k=10) BEST	0.2057	0.2569	0.3330

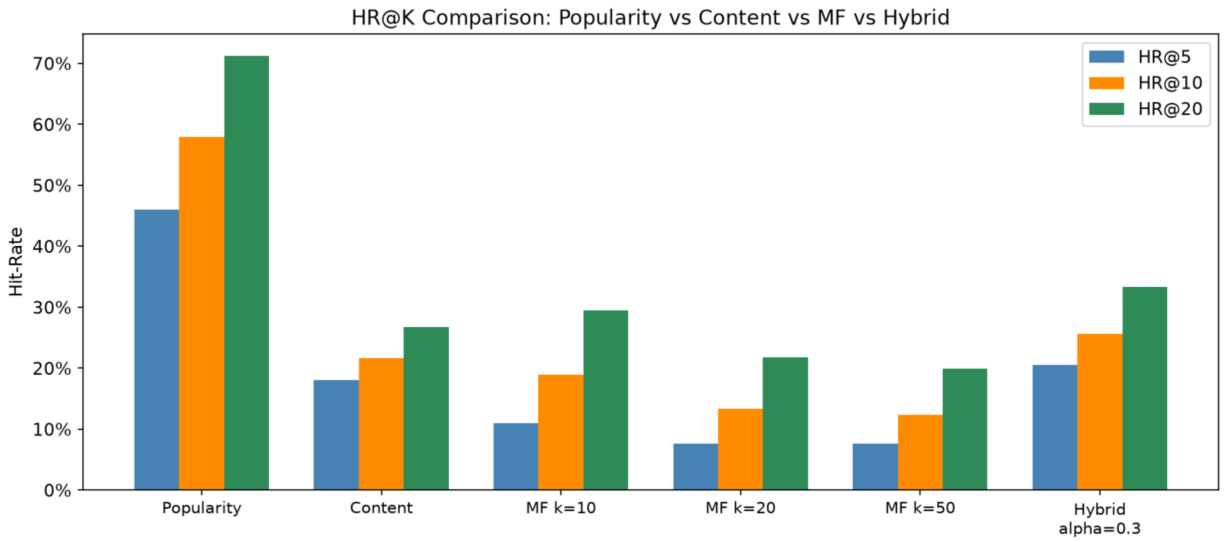

Figure: HR@K comparison. Hybrid (alpha=0.3) achieves best personalized HR@10=25.7%.

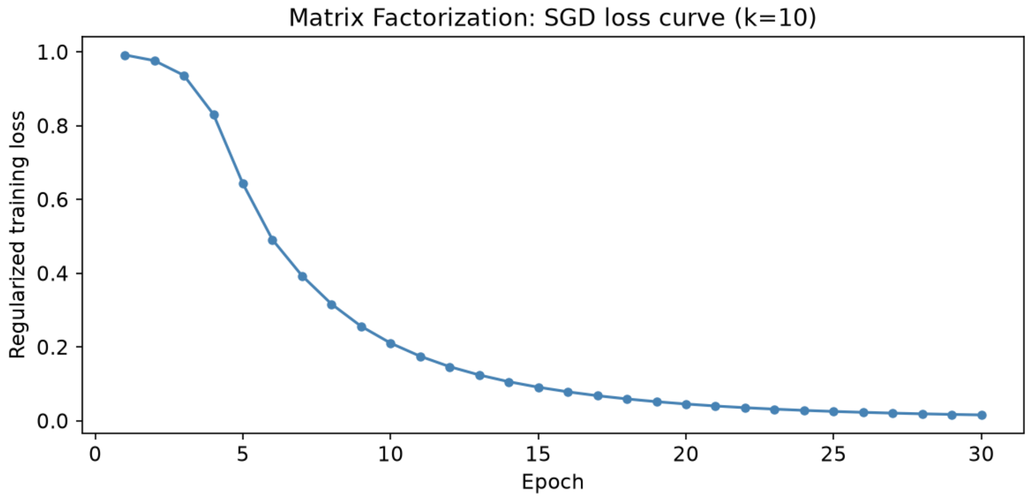

Figure: SGD loss per epoch (k=10). Monotonic decrease confirms convergence within 30 epochs.

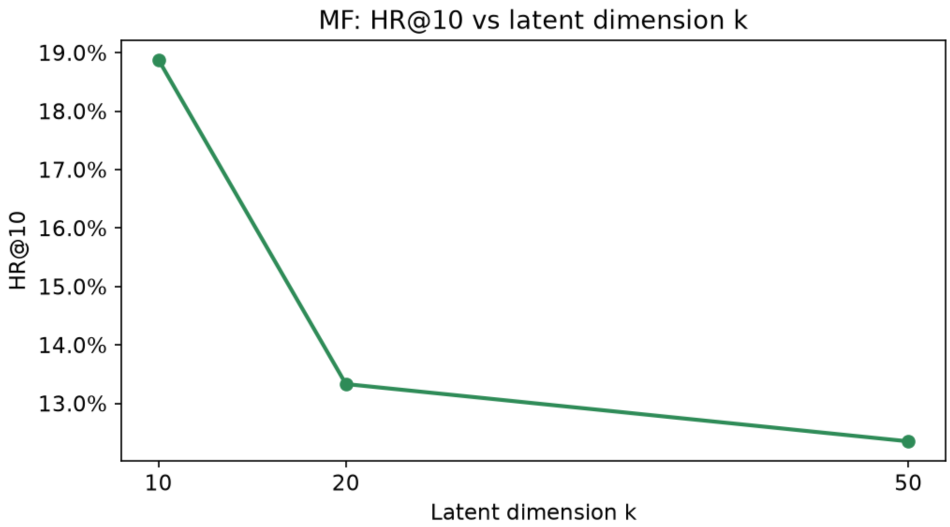

Figure: HR@10 vs latent dimension k. k=10 generalizes best on this sparse 484-game universe.

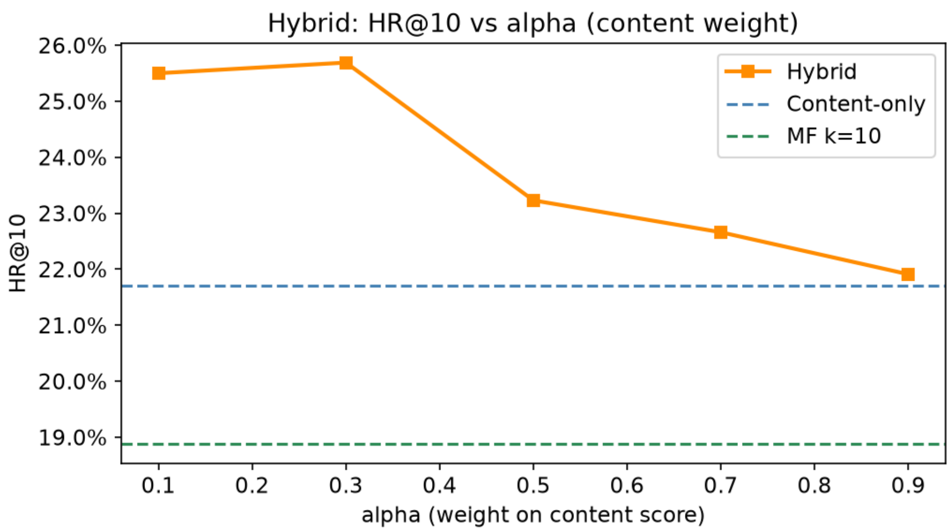

Figure: Hybrid HR@10 vs alpha. Optimal alpha=0.3 (30% content + 70% MF).

9. Error Analysis

Strong cases (hits): Users with history concentrated in one cluster get accurate recs. MF identifies similar narrow-profile users.

Failure cases (misses): Cross-cluster taste - history spans clusters (e.g., Action + Narrative) but held-out game belongs to a third cluster. Latent factors average out competing signals.

Secondary failure: Popularity contamination in users with only 2 qualifying interactions.

10. Reproducible Pipeline

```
python src/ingest.py              # Step 1: data ingestion
python src/feature_engineering.py # Step 2: feature matrices
python src/clustering_week7.py    # Step 3: clustering (Week 7)
python src/recommendation.py      # Step 4: recommendation engine
python src/generate_figures.py    # Step 5: report figures
```

All outputs deterministic with random_state=42.

11. Ethics and Access Note

- CC0 license. No PII beyond public Steam IDs used as opaque row indices.
- author_steamid: opaque identifier. No real names, emails, or financial data.
- Recommendation fairness: popularity bias present. Future: inverse propensity scoring.